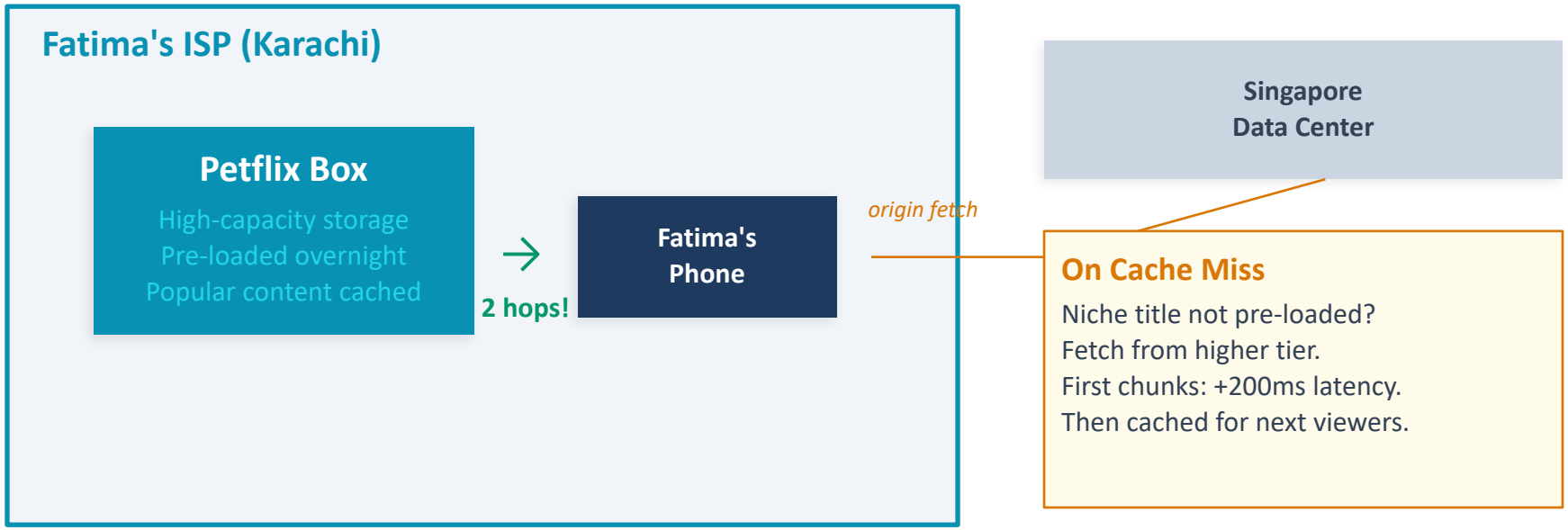
Layer 5

CDN — The Petflix Box

8,000+ boxes installed inside ISPs worldwide

Petflix Box: Video Served From Your ISP

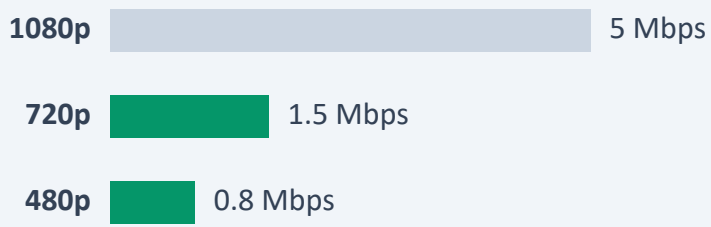


Win-win: Petflix ships boxes for free. ISP saves bandwidth. Users get low-latency HD video, even with unreliable international connectivity.

Per-Title Encoding & Adaptive Bitrate

"Mittens Goes to the Vet"

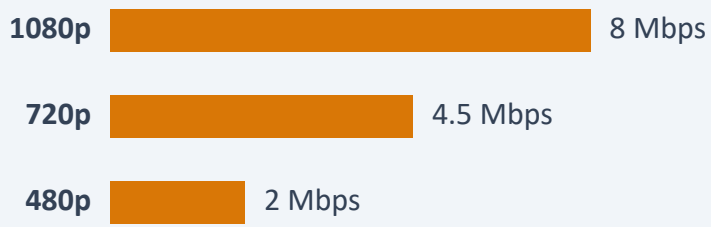
Low complexity: static shot, cat in carrier



720p looks identical to 1080p → encoded lower

"Dog Agility Championship"

High complexity: fast motion, fine detail



Needs high bitrates to look good (VMAF)

Adaptive Bitrate Streaming

Player monitors bandwidth and switches at chunk boundaries — seamlessly

Layer 6

Pods, Leader Election & Replication

How 5 machines agree on one truth

Leader-Follower Replication

Why Replicate?

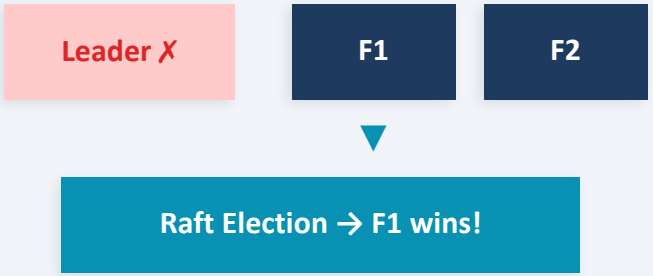
- Single machine = single point of failure
- 5 replicas: any one can die safely
- Trade-off: write latency for durability

The Consensus Problem

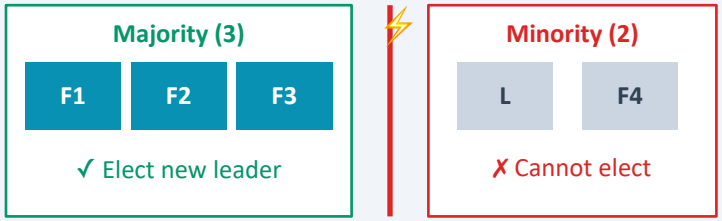
- At most one leader at any time
- Committed writes durable across majority
- Raft protocol (simpler than Paxos)

Leader Election & Network Partitions

Scenario: Leader Dies



Scenario: Network Partition



Raft Consensus Rules

Strict majority required

A candidate needs ≥3 of 5 votes. Only one majority exists in any group of 5.

Randomized timers

Followers start elections at random intervals to avoid split votes.

Old leader steps down

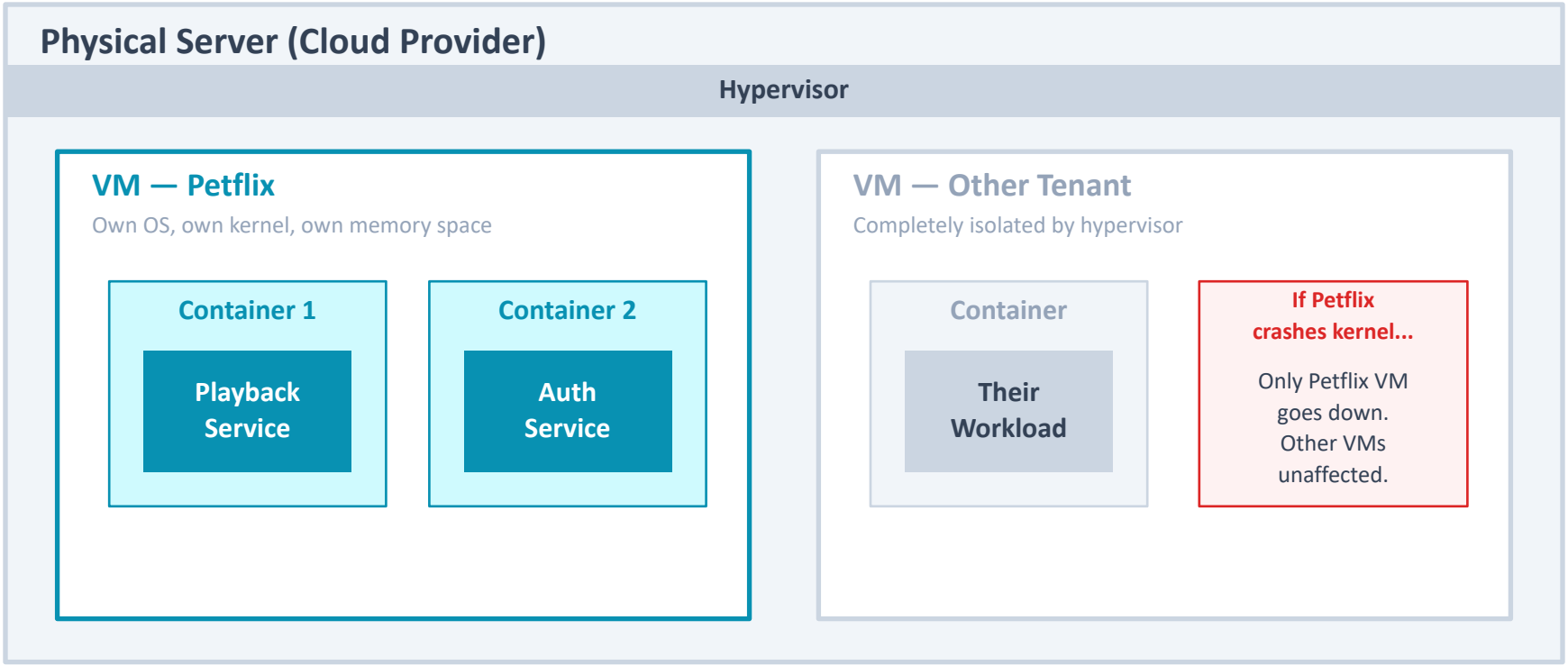
When isolated leader can't reach majority, it rejoins as follower after healing.

Layer 7

VMs, Containers & the Orchestrator

Russian nesting dolls of isolation

The Isolation Stack



VMs = strong isolation (own kernel) | Containers = lightweight isolation (shared kernel) | Both = defense in depth

Kubernetes: The Orchestrator

Schedule

Decide which containers run on which VMs

Scale

Spin up/down replicas based on load

Heal

Restart crashed containers automatically

Deploy

Rolling updates, one at a time, zero downtime

"Paws of Fury" Traffic Spike

Normal Load

Pod 1 | Pod 2 | Pod 3

Spike Detected → Auto-Scale

Pod 1 | Pod 2 | Pod 3 | New 1 | New 2 | New 3 | New 4

Noisy Neighbor Problem

Shared physical hardware means another tenant's I/O burst can slow you down.

Mitigation: resource reservation, or dedicated hardware for critical services.

Horizontal vs. Vertical scaling